Data Communications Planning – Student Handout

Lesson overview

By the end of this session you will be able to build a complete data communications plan for a signal operation, using the Philippine Army signal planning process. The session runs from requirements, to technology, to the plan, to voice-on-data, and ends with a group case scenario.

Terminal learning objective (TLO). At the end of this session, the students will be able to develop comprehensive data communications plans during signal operations.

No.	Enabling learning objective	Level	Where in this handout
8.4.1	Define data transmission requirements	C2	Part 2
8.4.2	Discuss data communication technologies	C3	Part 3
8.4.3	Apply data communications plans	C3	Part 4
8.4.4	Determine data networks in voice communication planning	C3	Part 5

Primary references.

- PAM 6-0102, Signal Operations (2022): planning process, signal estimate, signal plan.
- PAM 6-00, Command and Control Communications and Cyber Systems (C4S) (2017): architecture and connectivity rules.
- PAM 3-00, Army Operations: information management and information operations.

How to read this handout. Paragraph numbers such as *(0312)* refer to PAM 6-0102 unless a manual is named. Boxes marked **Supplementary** hold general technical knowledge that the manuals do not state. Check it against your equipment manuals and local SOP before relying on it in a plan.

Session flow.

1. Foundations (Part 1)
2. Requirements, technologies, plan, voice-on-data (Parts 2 to 5)
3. Check on learning (review questions, Annex E)
4. Case scenario: group planning and presentation (Part 6)

Part 1. Foundations

A data communications plan exists to keep the commander connected to every unit and every information source the mission needs, in the face of terrain, weather, the enemy and our own limits. It is part of the signal plan, not a separate IT project.

1.1 Signal and C4S

- **Signal** is the action that supports Army operations. **C4S** (Command and Control Communications, and Cyber Systems) is the set of capabilities that enables signal (PAM 6-00 0127).
- C4S has four sub-systems: communications, electronics, information and cyber (PAM 6-00 0401).
- The Army Signal Regiment (ASR) is the sole provider of signal support. Its NETBn runs information network operations, and its CyBn runs defensive cyber operations (0110, 0124).
- Signal support has four service categories: command and control communication, cyberspace operations and electronic warfare, repair and maintenance, and recovery and restoration (0131).

1.2 The three operating environments (0115 to 0118)

Environment	What it means for the data plan
Physical	An archipelago with distances over water, mountains that block line of sight, foliage that weakens propagation and satellite links, buildings and power lines in cities, and yearly typhoons that disable equipment.
Cyberspace	Hardware and transmission facilities (physical domain), configuration such as firewalls, topology and protocols (logical domain), and users (people domain).
Electromagnetic spectrum	Tactical radio, microwave, cellular and wireless networks, shared with the enemy. Control of the spectrum is the aim of electronic warfare.

1.3 Threats the plan must answer (0119 to 0123)

- Information warfare, such as cyberattack and electronic warfare used together.
- Hybrid threats: regular, irregular, terrorist and criminal elements acting together.
- Cyber threats against confidentiality, integrity or availability of networks.
- Insider threats.

Lessons from past operations (0111 to 0114): the Zamboanga siege (urban signal support), Typhoon Yolanda 2013 (commercial networks failed, Army tactical radio reported first), the Marawi siege (drones, digital maps, blue force tracking, live feeds), and COVID-19 (video-teleconferencing and information systems).

1.4 The Connect, Sustain, Recover framework (0144)

- **Connect:** connectivity for all Army units.
- **Sustain:** keep the connectivity going through the operation.
- **Recover:** repair and restore it when attacked or destroyed.

A good data communications plan answers all three questions for every link.

1.5 Principles that govern every plan

Principle	Source	In practice
Operations-focused	0209	Communications must never be the limiting factor on the commander's mission.
Interoperable	0210	Units within the Army and other services can talk to each other.
Redundant	0211	Multiple paths, backups, self-healing, data replication.
Scalable	0212	The network expands or shrinks by phase.
Secured	0213	COMSEC policy governs every service.
Reliability first	PAM 6-00 0201e	Reliability is paramount. Speed can outweigh security depending on conditions.
Survivability	PAM 6-00 0202a	Losing one part must not disrupt the whole system.
Appropriate technology	PAM 6-00 0203	The newest technology is not necessarily the best choice.

Part 2. Data transmission requirements (8.4.1)

A data transmission requirement is a statement of who must exchange what information, with whom, how quickly, how securely and over what terrain. Requirements come from the mission and the commander, not from the equipment on hand. Equipment is considered later, as a resource.

2.1 Where requirements come from (0305)

Source	What the signal planner takes from it
Commander's guidance or intent	The concept of operations, the means directed, and the priorities of establishment. The planner offers courses of action, and the commander decides.
Staff guidance	The needs of the users of signal support: operations, intelligence, logistics, civil-military operations.
Higher, subordinate and adjacent units	What the higher headquarters will provide, and what subordinate and adjacent units expect.
SOP	Installed capabilities and standing procedures, which shorten planning time.
CEOI and CESI	The CEOI says what to do (operating instructions for C4 systems). The CESI explains how to use the items

Source	What the signal planner takes from it
	in the CEOI.

2.2 The signal estimate and METAL (0302, 0303)

The signal estimate is an orderly examination of every factor, known or assumed, that may affect the signal mission. It is revised as the situation changes. The mnemonic **METAL** structures it:

- **M**ission, in precise terms.
- **E**nemy signal capabilities.
- **T**roops available to accomplish the mission.
- **A**rea of operations.
- **L**ogistical requirements.

The full format is in Annex A.

2.3 What the commander needs from the network

PAM 6-00 (0211) lists seven command and control requirements the infrastructure must support. Each one implies a data requirement.

C2 requirement	Data implied
Common operational picture	Situational display, messaging and database access shared to the lowest tactical unit.
Collaborative planning	Electronic exchange of plans between planners at separate locations.
Centralized command, decentralized execution	Fast distribution of directives and vital information.
Intelligence and operations integration	Prompt receipt and exchange of intelligence.
Apportionment of resources	Current status and location of resources, and directives to move them.
Information push and pull	Delivery to superiors and subordinates, and access to reliable sources.
Security	Information goes only to those who need it, and the network withstands cyberattack.

PAM 3-00 (0847) adds six qualities that information must have: **accurate, timely, usable, complete, precise, reliable**. PAM 3-00 (0849) adds that commanders express their needs through **CCIR**, which generate PIR, FFIR and EEFI. Information that arrives late, or in a form no one can read, has the same effect as none (0839).

2.4 Requirement categories

Use these headings to state each requirement. The first five come from the principles in Part 1. The last three come from the operating environment.

Category	Question the planner asks
Reliability and redundancy	What is the minimum acceptable availability, and what is the backup path?
Security	What classification, and what COMSEC and cyber protection does the data need?
Speed and capacity	How much data, how often, and how fast must it arrive?
Interoperability	Which units, services, agencies or allies must be reachable?
Scalability	How will the user count change by phase?
Coverage and range	Which terrain, islands and distances must the link cross?
Mobility	Is the user on the move, on the halt or fixed?
Environmental resilience and sustainment	Can the equipment survive weather, power loss and rough handling, and who maintains it?

> ***Supplementary: quantifying a requirement.*** *The manuals do not give a method for sizing traffic. A workable approach is to list each information exchange requirement with these fields: originator, recipient, information type (voice, text message, image, video, database, map or track data), frequency and volume, maximum acceptable delay, precedence, classification, and the phase when it applies. Then total the volume by link. Use the Annex C worksheet.*

Example of one filled-in requirement

Originator	Recipient	Information	Volume and frequency	Max delay	Precedence	Classification	Phase
Maneuver battalion TOC	Brigade TOC	Position reports and blue force track	Small text records, every few minutes	Under 1 minute	Priority	Restricted	Execution
Brigade TOC	Division TOC	Drone video feed	Continuous video	Real time	Immediate	Confidential	Execution

2.5 Check: do the requirements pass the test?

Before you use a requirement, check that it is specific, measurable, tied to the mission or the commander's CCIR, ranked in priority, and phased. Rank them, because resources are always limited (0310).

Part 3. Data communication technologies (8.4.2)

The planner picks the technology that meets the requirement in the terrain, spectrum and threat of the operation. No single technology works everywhere, so every plan combines several. PAM 6-0102 groups the Army's means into fixed, tactical and special purpose communications.

3.1 Fixed communications (0133, PAM 6-00 0403)

- **Role:** the permanent backbone for voice and data between camps and bases.
- **BONTEX** (Bonifacio Telephone Exchange): telephone service linking Army units to HPA and GHQ. The Army is shifting to IP telephony.
- **PANET** (Philippine Army Network): the secured, managed intranet carrying administrative and mission-critical information systems. Its primary hub is at the PA communications center, Fort Bonifacio (PAM 6-00 0414).
- **AFP Fixed Communications System (AFP FCS):** the transmission "superhighway" between switches, run by GHQ-AFP.
- **DBTOCS** (deployable brigade tactical operations center system): extends fixed communications down to brigades.
- **Components:** transmission, switching, cable, terminal and support facilities (power, buildings, environmental control).
- **Services:** telephone, video-teleconferencing, email and collaboration, and information systems.

3.2 Tactical communications (0134)

- **Role:** communications on the move and on the halt, over the radio frequency spectrum.
- **Radios:** HF, VHF and UHF. The Army fielded HF/SSB 20-watt manpack and VHF/FM 5-watt handheld radios under the modernization program (PAM 6-00 0514).
- **Networks:** the combat net radio (CNR) system, using radio-over-IP (RoIP) and ad-hoc networks.
- **Non-military radios** complement tactical radios, mainly to reach other agencies in emergencies and disasters.
- Voice remains the bedrock of tactical communication.

3.3 Special purpose communications (0135, PAM 6-00 0405)

- **Role:** to reach places the fixed system cannot, and to augment it with commercial services.
- **Means:** VSAT satellite terminals, mobile satellite communications, commercial cellular, satellite phones and commercial video-conferencing platforms.
- **Control:** use "with utmost prudence", under COMSEC policy.

3.4 Which means for which link (PAM 6-00 0406)

The manual sets a primary, secondary and tertiary means for each link between command levels. Use it as your default routing plan and justify any change.

Link	Primary	Secondary	Tertiary or other
Within strategic HQs	AFP FCS	HF/SSB	Commercial landline. Cellular while on the move.
Strategic to operational HQ	AFP FCS, with PA drop-and-insert to major camps	HF/SSB	Commercial landline and mobile, if funded
Operational to tactical HQ	AFP FCS if technically feasible	HF/SSB with VHF/FM or UHF/FM, linked through repeaters	Commercial landline or mobile
Tactical HQ to operating units	AFP FCS if feasible	HF/SSB, with VHF/FM or UHF/FM	Commercial landline or mobile
Unit HQ to unit elements	Standardized mobile systems	Not stated	Not stated

3.5 How the environment drives the choice (0116)

Condition	Effect	Planning response
Islands and open water	Line of sight fails over distance	HF, satellite or relay stations
Mountains	Block line-of-sight radio	Repeaters on high ground, a propagation study
Dense foliage	Weakens propagation, including satellite links	Site antennas with clear sky, test before committing
Cities	Buildings, towers and power lines distort radio	Short hops, wired or cellular backup
Typhoons	Disable commercial and Army assets	Deployable signal teams, mobile command center, tactical radio as backup (Yolanda lesson)
Enemy EW	Jamming and interception	Frequency hopping, anti-jamming, procedures for changing frequencies

3.6 Spectrum and protection

- **Spectrum management** is planning, coordinating and managing the use of the spectrum. It must comply with National Telecommunications Commission (NTC) rules (0218).
- **Electronic warfare** has three objectives: support, attack and protect (0138, 0223). Protection includes frequency hopping, anti-jamming and counter-countermeasures.

- **COMSEC policy** governs every service. The Vigilance Net is less secure, so it relies on ciphertext and double talk (0316).

3.7 Supplementary: comparing the means

> ***Supplementary.*** *The manuals describe the technologies but do not compare them. This table is general knowledge for orientation. Replace it with the figures in your equipment manuals when you plan.*

Means	Strength	Limitation	Typical data use
HF/SSB radio	Long range without infrastructure, skywave beyond line of sight	Low data rate, affected by ionospheric conditions	Short messages, position reports, backup voice
VHF/UHF radio	Reliable over line of sight, small handhelds	Terrain-limited, range depends on relays	Tactical voice, short messages, RoIP bridging
Satellite, geostationary (VSAT)	Wide coverage, high capacity, independent of terrain	Higher delay, needs clear sky and a power supply, rain and foliage reduce the link	Backbone links, video, bulk data to remote sites
Satellite, low earth orbit	Lower delay than geostationary	Needs terminals and service arrangements, supply and security need review	Remote-site data and voice
Commercial cellular	Wide urban coverage, cheap	Fails in disasters, not secure, exposed to interception	Routine or administrative traffic, never classified
Fiber and microwave	Very high capacity, reliable when intact	Fixed, vulnerable to cuts and weather, slow to establish	Camp-to-camp backbone, PANET
Mobile ad-hoc network (MANET)	Self-forming, extends coverage, tolerates node loss	Capacity falls as hops increase, needs power and monitoring	Dismounted and vehicle data, blue force tracking

3.8 Supplementary: network vocabulary

> ***Supplementary.*** *These basics help you read a network diagram and discuss it with network engineers. Each term is tied to its Army use.*

Term	Plain meaning	Army example
LAN	Network in one building or camp	The network inside a TOC
WAN	Network linking LANs over	PANET between camps, over the AFP FCS or

Term	Plain meaning	Army example
	long distances	satellite
VPN	Encrypted tunnel over a shared network	Protected link over a commercial circuit (check approval and COMSEC policy first)
Router	Directs packets between networks	Connects a unit LAN to PANET
Switch	Connects devices within a LAN	Connects computers and IP phones inside a TOC
Firewall	Filters traffic by rules	Protects PANET from outside networks
Star topology	Everything connects to one center	Simple but the center is a single point of failure
Mesh topology	Many paths between nodes	Survives loss of a node, suits survivability and MANET
Hybrid topology	Mix of the above	Typical of a real Army network: a star at the TOC, a mesh in the field

Part 4. The data communications plan (8.4.3)

The data communications plan is built inside the signal plan and published as the C4S annex to the operations order (0314). It follows the signal planning process, which runs alongside the military decision-making process, or troop leading procedures for small units (0304). Your task in this part is to apply the process, not only to describe it.

4.1 The signal planning process (0304 to 0314)

1. **Develop signal and C4S requirements (0305).** Gather them from the commander's intent, staff, higher, adjacent and subordinate units, SOP, and the CEOI and CESI. This is Part 2 of this handout.
2. **Determine available resources (0306 to 0309).** Keep a current chart of three things:
    - Equipment: authorized against on hand, serviceable against unserviceable, and reserve.
    - Personnel: authorized against actual fill, training status, critical shortages, committed or deployed, and reserve.
    - Funds: installation, operations and maintenance of C4 systems.
3. **Develop priorities (0310).** Compare requirements to capabilities. Where resources fall short, rank support by supportability and importance to the mission.
4. **Develop recovery procedures (0311).** Plan for interrupted connectivity, electronic warfare, cyberattack and compromised frequencies (section 4.4).
5. **Prepare a tentative plan (0312).** Write the concept of support, allocate resources, site the C4S centers, set security measures, and design interconnections and alternate routing (section 4.2).
6. **Finalize the plan (0313).** Verify it against the checklist in section 4.3.

7. **Publish plans and orders (0314).** Prepare the signal plan or C4S annex, with appendices and tabs, and give the operations officer the signal estimate for inclusion in the order.

4.2 What the tentative plan must contain (0312)

- **General concept of signal support** with a **primary, alternate, contingency and emergency** plan. This is the PACE idea: for every critical service, decide the main means and three fallbacks before the operation begins.
- **Allocation of resources** to requirements.
- **Siting of C4S centers and facilities**, based on requirement density, tactical disposition of units, location of support units, terrain, propagation, projected operations, accessibility, and administrative and logistics needs.
- **Security measures.**
- **A general plan of interconnections and alternate routing.**

PACE example from the Annex B sample plan: primary Harris handheld VHF tactical radio, alternate Harris manpack tactical radio, emergency messenger. Notice that a data plan needs the same four-level thinking for each data service, not only for voice.

4.3 Finalization checklist (0313)

- The plan meets the command's signal support requirements.
- It is practical, and the resources and capabilities are adequate.
- Its emphasis agrees with the priorities set in step 3.
- Areas of responsibility are assigned for each C4S center and facility.
- A site survey or reconnaissance and a **propagation study** were made.
- Center and facility locations are pinpointed.
- Signal **coverage and gaps** are identified.
- Security measures are in place in the operating environment.

4.4 Recovery procedures (0311)

Contingency	The plan must include
Interrupted connectivity (configuration faults or damaged equipment)	Redundant communications, equipment repair and replacement, and on-site technical support
Electronic warfare (jamming and interception)	Countermeasures and counter-countermeasures
Cyberattack	Monitoring of the PA network including the mobile ad-hoc network, and procedures for the CSIRT and first responders
Compromised frequencies (captured equipment)	Procedures to change frequencies efficiently, in the shortest possible time

The four deployable signal teams that execute recovery (0143, 0228 to 0231): the TOC team (TOC deployment plan, mobile command center), the C3 restoration team, the network recovery team, and the cyber security incident response team (incident response plan).

4.5 Signal plan format (Annex B)

The signal plan follows the operation order format. Its main parts:

1. Situation: enemy forces, friendly forces, attachments and detachments.
2. Mission.
3. Execution: concept of operations, concept of support by phase (each with task, purpose, method and effect), tasks to maneuver and support units, C-E support, coordinating instructions (including the primary, alternate and emergency means).
4. Sustainment.
5. Command and signal.

The sample plan uses three phases: planning and preparation, execution, and consolidation and reorganization. Annex B gives the layout.

4.6 Suggested data appendix

> ***Supplementary.*** *The Annex B format has an empty "C-E Support" paragraph and no data-specific tab. A usable practice is to attach a data communications appendix to the signal plan. Annex D is a template. It should contain a network diagram, a service list, a link and routing table with PACE for each service, an addressing and naming scheme, the frequency and channel plan (or a reference to the CEOI), the security plan (COMSEC and cyber), network operations and monitoring, and the recovery plan. Follow your unit SOP where it differs.*

4.7 Special operating conditions

Condition	Planning points
Joint operations (0315 to 0319)	Interoperability with other services is the key. Options: use the Vigilance Net (with ciphertext and double talk), lend equipment to another service, embed Army radio operators with it, or embed its operators at the Army TOC closest to the operating units (the most common practice).
Coalition operations (0320 to 0322)	Each nation keeps its own sovereign network. Identify the points of integration where networks connect.
Humanitarian assistance and disaster response (0323 to 0326)	Follow the NDRRM plan and the incident command system. The HADR radio net must be separate from the secured operational net and is treated as unclassified and unsecured. Support also goes to other stakeholders in the response. Deployable signal teams and the mobile command center are put on standby.

PAM 6-00 (0213) also expects a communications plan and CEOI before a disaster, and redundant communications during the response.

Part 5. Data networks in voice communication planning (8.4.4)

Voice is still the bedrock of tactical communication (0134), but more and more of it now travels as data. Once voice rides on a data network, a voice plan is only as reliable as the network under it. The planner must therefore decide which voice services depend on data networks, size those networks for voice, and keep a voice path that survives their failure.

5.1 Where voice meets data in the Army

Service	How it uses the data network	Manual reference
Fixed telephone	BONTEX is being replaced by IP telephony (voice over IP) over PANET.	0133
Combat net radio	The CNR system uses radio-over-IP (RoIP) and ad-hoc network technologies, so radio nets can be linked across the network.	0134
Secure voice and video-teleconferencing	Supported at the AOC and TOCs of major units, over the C4ISTAR network.	PAM 6-00 0505, 0506
Commercial video-conferencing, cellular and satellite phones	Special purpose services, used with care under COMSEC policy.	0135
Voice, data, image and video	The communications infrastructure must support all four forms of information.	PAM 6-00 0402

5.2 What a voice-on-data plan must determine

1. **Which voice services depend on the data network.** List each voice service and the network or link it relies on.
2. **The capacity the voice needs.** Size each link for the number of simultaneous calls plus data traffic (see 5.3).
3. **Priority.** Voice and command traffic must not be starved by bulk data. The plan needs a priority rule.
4. **The fallback path.** Each critical voice service needs a primary, alternate, contingency and emergency means that does not share the same failure point (section 4.2). Use the PA default: AFP FCS, then HF/SSB, then VHF/UHF, then commercial (PAM 6-00 0406).
5. **Security.** Secure voice stays on the secured network. Commercial and Vigilance Net voice is treated as exposed, so use ciphertext and double talk where it applies (0316). The HADR net is separate and unsecured (0324).

6. **Dependencies.** Power, antennas, switches and the network operations center must also survive. A phone that works only while the router has power is a single point of failure.
7. **Recovery.** Name the team and the procedure that restores voice when the network fails (section 4.4).

5.3 Supplementary: sizing voice on a link

> ***Supplementary.*** *The manuals give no sizing method. These are rough planning figures only. Confirm with your equipment documentation.*

A single IP voice call needs bandwidth in both directions. Roughly:

Voice coding	Approximate bandwidth per call, including network overhead
Uncompressed (G.711)	About 80 to 90 kbps
Compressed (G.729)	About 25 to 30 kbps

A satellite or radio link carrying voice and data needs a compressed codec and a priority rule. Satellite links also add delay, which affects conversation quality more than raw capacity does.

Worked example. A brigade TOC expects 15 simultaneous compressed calls to division over a VSAT link, plus 400 kbps of other data. Voice is about 15 × 30 kbps = 450 kbps. Add the data (400 kbps), and a margin of at least 25 percent for peaks and overhead: (450 + 400) × 1.25 ≈ 1,060 kbps. Plan the link at roughly 1.1 Mbps or more, or reduce the call count, or move bulk data off the link at peak hours.

5.4 Planning table for one voice service

Use this pattern for each voice service in your plan.

Service	Depends on	Primary	Alternate	Contingency	Emergency	Capacity	Security
Brigade TOC to division voice	PANET, VSAT terminal, power	IP telephony over PANET via AFP FCS	VSAT voice	HF/SSB voice	Messenger	About 450 kbps for 15 calls	Secured network, COMSEC applies

5.5 Common planning errors

- Treating IP telephony and the data network as separate plans, so the voice path has no data backup.
- Using one commercial circuit for both the primary and the alternate path.
- Sizing the link for data only, so voice quality fails at peak.
- Connecting the HADR net to the secured net.

Part 6. Case scenario: Task Force SENTINEL

Your group is the signal staff of Task Force SENTINEL. You must produce a data communications plan for a two-phase operation that turns from a security operation into disaster response. This scenario tests the terminal learning objective and all four enabling objectives.

All units, places and names in this scenario are fictional. Where the scenario is silent, state a reasonable assumption in your plan and mark it as an assumption.

6.1 Situation

Area of operations. Two islands of the fictional province of Verdana. **Isla Mayor** (about 60 km north to south) has a coastal lowland with towns and cellular coverage, and a forested mountain range inland. **Isla Menor** lies 35 km across open water to the east and is hilly, with little commercial coverage. The area is typhoon-prone, and the typhoon season has started.

Enemy. The Verdana Armed Group (VAG) operates in the Isla Mayor mountains and has cells on Isla Menor. Intelligence assesses that the VAG:

- monitors commercial cellular traffic and unencrypted VHF with handheld scanners;
- may jam local VHF channels at close range;
- uses social media for information operations and probes Army and local government networks;
- may capture radios in ambush.

Friendly forces.

Element	Location	Notes
Task Force SENTINEL TOC	Camp Sinag, Isla Mayor lowlands	Fixed facility with a PANET connection through one microwave link to the AFP FCS
Forward tactical command post (TCP)	Mountain foothills, 25 km from the TOC	Mobile, set up on the halt
1st Infantry Battalion	Isla Mayor mountains	Operates dismounted, some companies beyond line of sight of the TCP
2nd Infantry Battalion	Isla Menor	Reaches the TOC only over water
Higher headquarters	Division HQ, outside the area	Reached through PANET and AFP FCS
Air and naval support	Philippine Air Force and Philippine Navy elements	Under their own networks. Liaison is needed for joint coordination
Local government and responding agencies	Provincial and municipal offices	Involved in the disaster response phase. They use commercial radios and cellular

Mission. Task Force SENTINEL conducts security operations in Verdana from D-day to D+10 to neutralize the VAG, and provides support to disaster response operations as directed.

Concept in two phases.

- **Phase 1, D-3 to D+3: security operation.** Battalions move into the mountains and Isla Menor. The TOC needs a common operational picture, blue force tracking, drone video from both battalions, and daily reporting to division.
- **Phase 2, D+4 to D+10: typhoon response and continued security.** A typhoon makes landfall on D+4. The task force is directed to support disaster response while keeping pressure on the VAG.

6.2 Signal assets and constraints

Item	Quantity and notes
HF/SSB manpack radios	12
VHF/FM handheld radios	60. Some are encryption-capable, and the rest are clear voice only
UHF commercial radios	20
Repeater sets	2
VSAT terminal	1, assigned to the TOC
Mobile satellite terminal	1
Deployable brigade tactical operations center kit	1
Mobile command center	1
Satellite phones	4
Commercial cellular	Available in the Isla Mayor lowlands only
Signal platoon	1 officer and 28 enlisted personnel. No qualified network administrator. 1 radio technician
Maintenance	A signal mobile maintenance and repair team (SMMART) is available on request
Funds	Enough for routine commercial service and consumables. Anything more must be requested through the G6

Power is available at the TOC. Field units depend on batteries and vehicle generators.

6.3 Injects

The instructor releases these during planning or briefing. Your plan must show how it responds to each one.

Inject	Event
1 (D+1)	A 1st Battalion company is ambushed. Two handheld radios are captured. Some

Inject	Event
	were loaded with the operational frequencies.
2 (D+3)	Division asks the TF to send drone video and a daily briefing by video-teleconference at 0600 each day.
3 (D+4)	The typhoon cuts the microwave link to the AFP FCS and takes down commercial cellular on Isla Mayor. The TF is directed to support disaster response.
4 (D+5)	The cyber monitoring cell reports suspicious traffic and a possible malware infection on a TOC laptop connected to PANET.
5 (D+6)	A navy patrol boat and provincial disaster officials ask to join the task force radio net.

6.4 Your tasks

Produce the following as a group:

1. **Signal estimate (condensed).** Use the Annex A headings and the METAL mnemonic. Maximum two pages.
2. **Signal plan with a data communications appendix.** Use the Annex B layout and the Annex C and D templates. The plan must contain:
    - at least eight information exchange requirements, with a priority for each (8.4.1);
    - the technology selected for each link, with the reason, tied to the terrain, spectrum, threat and the PA connectivity hierarchy (8.4.2);
    - a PACE table for at least five services, including at least two voice services (8.4.3, 8.4.4);
    - a simple network diagram showing the TOC, TCP, both battalions, division and the other agencies;
    - a sizing calculation for the TOC-to-division satellite link, with voice, video-teleconference and data (8.4.4);
    - security measures, including how the disaster response net is kept separate;
    - your response to each of the five injects, with the recovery team that acts.
3. **Brief.** Ten minutes to the commander (instructor), then five minutes of questions.

Suggested time. Planning and drafting 90 minutes, briefings 15 minutes per group, discussion 20 minutes. The instructor may adjust.

6.5 Evaluation

Criterion	What the evaluator looks for	Objective	Points
Requirements	Eight or more clear requirements, linked to the mission and CCIR, with volume, timeliness, classification and priority	8.4.1	20
Technology selection	Each means fits terrain, threat and the connectivity hierarchy. Reasons are stated. Limits are	8.4.2	15

Criterion	What the evaluator looks for	Objective	Points
	acknowledged		
Plan structure and process	Follows the planning process and format. Includes resources, priorities, finalization checks and an assumptions list	8.4.3	20
Redundancy and recovery	PACE for each critical service with independent paths. All five injects handled. The right recovery team is named	8.4.3	15
Voice on data	Voice dependencies on the network shown. Sizing is correct and shows the reasoning. Voice fallback is independent	8.4.4	15
Security and separation	COMSEC, cyber, compromised frequencies and the separate disaster response net are addressed	8.4.2, 8.4.3	10
Brief and teamwork	Clear, within time, and answers questions with evidence from the plan	TLO	5
Total			**100**

A group that earns at least 80 points, with no criterion below half its points, has demonstrated the terminal learning objective.

Annexes

Annex A. Signal estimate format (PAM 6-0102 Annex A)

1. **Mission**
2. **Situation and courses of action**
    - Area of operations and area of interest analysis: terrain, weather, existing communications
    - Enemy: organizational structure, disposition and location, strengths and weaknesses, capabilities and limitations
    - Own situation: organization of forces, disposition and location, personnel and training status (assumed), inventory of signal assets (assumed), capabilities
    - Courses of action
3. **Analysis of opposing courses of action**
    - Communications and disposition of opposing forces
    - Signal propagation of opposing forces
    - Enemy capabilities and countermeasures (request for information to intelligence): capabilities and limitations
    - Friendly capabilities and limitations
    - Countermeasures: against enemy capabilities, and for signal platoon limitations
4. **Comparison of courses of action**

5. **Decision and recommendations**
    - Personnel requirements
    - C4S requirements
    - Sustainment requirements
    - Means of communication
    - Location of the TCP and mobile command post
    - Task list (implied and specific), for the commander's decision

Annex B. Signal plan format (PAM 6-0102 Annex B)

- Heading: copy number, issuing unit, place, date-time group, message reference number, plan title and the order it supports, references, time zone, task organization.
- **1. Situation:** (a) enemy forces, (b) friendly forces, (c) attachments and detachments, (d) facts.
- **2. Mission.**
- **3. Execution:** (a) concept of operations; (b) concept of support by phase, each with task, purpose, method and effect (Phase 1 planning and preparation, Phase 2 execution, Phase 3 consolidation and reorganization); (c) tasks to maneuver units; (d) tasks to support units (signal, intelligence, operations, logistics); (e) C-E support; (f) coordinating instructions, including the primary, alternate and emergency means of communication.
- **4. Sustainment.**
- **5. Command and signal:** (a) command, (b) signal.
- Authentication block.

Annex C. Information exchange requirements worksheet

Copy this table and add rows. Add one row per requirement, then total the volume per link.

No.	Originator	Recipient	Information type (voice, text, image, video, data)	Volume and frequency	Max delay	Precedence	Classification	Phase	Link used
1									
2									
3									
4									

Annex D. Data communications appendix template and PACE worksheet

> ***Supplementary template.*** *Not part of Annex B. Follow your unit SOP where it differs.*

Appendix contents.

1. Purpose and assumptions
2. Network diagram (nodes, links, means, locations)
3. Services list (voice, messaging, common operational picture, video, email, database)

4. Link and routing table with PACE (below)
5. Addressing and naming, and the frequency and channel plan (or the CEOI reference)
6. Security: COMSEC, cyber, separation of unclassified and secured nets
7. Network operations and monitoring: who, where, when
8. Recovery: teams, triggers and procedures for each contingency
9. Sustainment: power, maintenance, spares

PACE worksheet. One row per service.

Service	Users	Primary	Alternate	Contingency	Emergency	Capacity needed	Security	Trigger to switch	Recovery team

Annex E. Review questions

Answer these before the case scenario. Cite the paragraph from the manual where one applies.

1. What does the mnemonic METAL stand for, and what is the signal estimate used for? (8.4.1)
2. Name four sources of data transmission requirements for the signal planner. (8.4.1)
3. Why does an information requirement need a maximum delay and a precedence, not only a volume? (8.4.1)
4. What are the three categories of signal communications in PAM 6-0102, and what does each include? (8.4.2)
5. For a link between an operational headquarters and a tactical headquarters, what are the primary and secondary means under PAM 6-00 0406? (8.4.2)
6. Name three effects of the Philippine physical environment on radio and satellite links, and one planning response for each. (8.4.2)
7. List the seven steps of the signal planning process. (8.4.3)
8. What must the plan include for each of the four contingencies in paragraph 0311? (8.4.3)
9. How do you keep the HADR radio net from compromising the secured operational net? (8.4.3)
10. A brigade TOC needs 20 simultaneous compressed voice calls and 500 kbps of data over one link. Estimate the link size with a 25 percent margin, and state your assumption about per-call bandwidth. (8.4.4)
11. Why is a voice service that runs over IP not made safe just by having a backup router? (8.4.4)
12. Write a PACE entry for secure voice between a battalion TOC and brigade TOC on an island beyond line of sight. (8.4.3, 8.4.4)

Annex F. Abbreviations

Term	Meaning
AFP FCS	Armed Forces of the Philippines Fixed Communications System

Term	Meaning
AOC	Army Operations Center
ASR	Army Signal Regiment
BONTEX	Bonifacio Telephone Exchange
C4S	Command and Control Communications, and Cyber Systems
CCIR	Commander's critical information requirements
CEOI	Communication-electronics operating instructions
CESI	Communication-electronics standing instructions
CNR	Combat net radio
COMSEC	Communications security
CSIRT	Cyber security incident response team
DBTOCS	Deployable brigade tactical operations center system
DST	Deployable signal team
EW	Electronic warfare
HADR	Humanitarian assistance and disaster response
MANET	Mobile ad-hoc network
MCC	Mobile command center
METAL	Mission, enemy, troops, area of operations, logistics
NTC	National Telecommunications Commission
PACE	Primary, alternate, contingency, emergency
PANET	Philippine Army Network
RoIP	Radio over Internet Protocol
SMMART	Signal mobile maintenance and repair team
TCP	Tactical command post
TOC	Tactical operations center
VSAT	Very small aperture terminal

Annex G. References

- Philippine Army Manual 6-0102, *Signal Operations*, Army Signal Regiment, Philippine Army, promulgated 15 April 2022.
- Philippine Army Manual 6-00, *Command and Control Communications, and Cyber Systems*, 2nd edition, 11 December 2017.
- Philippine Army Manual 3-00, *Army Operations*.
- The CEOI, CESI and unit SOP in force, and the equipment manuals for the systems in your plan.